Image Processing (EE 604)

Homework-1

Q1) Bit-Plane Slicing and Progressive Image Reconstruction

Using the bit-plane slicing technique, create a video that demonstrates the progressive reconstruction of a grayscale image by cumulatively adding its bit planes.

Start with **only the 8th (most significant) bit plane** of the image. Then, add the remaining bit planes **one at a time in the following order: 7th, 6th, 5th, 4th, 3rd, 2nd, and 1st**. Thus, the video should show the following sequence:

8th → 8th+7th → 8th+7th+6th → ... → 8th+7th+6th+5th+4th+3rd+2nd+1st

The final frame should therefore correspond to the **original image reconstructed using all eight bit planes**.

You may use any grayscale image for this demonstration. A reference video has been attached to help you understand the expected output.

Your write-up should include the code along with the comments, the 8 generated images and parameters such as frame rate. Please upload **the resultant video** demonstrating the progressive reconstruction.

Q2) Image Degradation and Restoration Using Spatial Filters

Start with a clean grayscale image and generate **at least four degraded versions** of the image by adding **four different noise models**. For each noise model:

1. Display the original clean image and the corresponding noisy/degraded image.
2. Apply **appropriate spatial filters** to the degraded image to restore it toward the original image. Display the restored image.
3. Compare the restored image with the original clean image using **PSNR and SSIM**.
4. Briefly explain why the selected spatial filter(s) are suitable for the corresponding noise model.